

Advait Rajesh Bharadwaj (he/him)

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EDUCATION

University of Massachusetts Amherst

Amherst, MA, USA

Manning College of Information & Computer Sciences and College of Natural Sciences

GPA: 3.681/4.00

Bachelor of Science in Computer Science and Mathematics (Statistics and Data Science)

Exp. Graduation, May 2026

- Courses:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Statistics, Linear Algebra, Discrete Mathematics, Probability, Multivariate Calculus

ACHIEVEMENTS

- Won “Best Generative AI Hack” at **HackHer(413) 2025** and “Best UMass Hack” at **HackUMass XII 2024**
- Secured **\$7,000** funding by winning the Executive Pitch Award for Roommatez, a roommate-matching startup
- Awarded **\$46,000** Chancellor’s Scholarship for academic excellence
- Dean’s List for five consecutive semesters (**2023–2025**)

EXPERIENCE

PropertyAngel Management Pvt. Ltd (LangChain, Firestore, Google Custom Search API, BeautifulSoup)

Bangalore, India

AI/ML Engineer

May 2025 — September 2025

- Led development of an AI-driven multi-agent system (**15+ GPT-4-powered bots**) to automate property workflows (rent/sale price analysis, property search, rental agreements).
- Designed session-based memory (Firestore) and integrated Google Custom Search API to serve real-time pricing data.
- Used LangChain for agent orchestration, OpenAI GPT-4o for LLM tasks, and FAISS for vector search over **10,000+** indexed properties in over **15 cities** across India.
- Built internal cost-optimization tools and prompt frameworks, reducing token usage by **~40%**.
- Deployed the solution using Docker and Google App Engine, enabling production-level stability with **sub-3s** average response times across **1000+** test sessions.

University of Massachusetts Amherst

Amherst, MA

Undergraduate Teaching Assistant for CS383: Artificial Intelligence

September 2024 — May 2025

- Assisted **300+** students in a core AI course by answering queries, providing feedback on assignments, and holding office hours, which improved student performance and engagement in complex topics like machine learning and search algorithms.

Undergraduate Research Assistant

May 2024 — September 2024

- Investigated classical vs. quantum machine learning models for genomic sequence classification (**100,000+ DNA sequences**) using Python (Scikit-learn) and IBM Qiskit.
- Analyzed model accuracy and demonstrated that state-of-the-art classical algorithms currently outperform quantum models, providing insight for future research directions.

PROJECTS

Math127GPT (React, Vite, TailwindCSS, FastAPI, LangChain, AWS S3, FAISS)

github/ShadowKrius/Math127GPT

- Developed an AI chatbot to help **2000+** students with Calculus problems by retrieving relevant content from course materials and using an LLM (OpenAI GPT-3.5) for step-by-step explanations.
- Engineered a full-stack solution with a React frontend and FastAPI backend; integrated a FAISS vector store for semantic search over textbook PDFs and utilized LangChain + OpenAI API to generate accurate, context-aware tutoring responses.

Resonance (Python; Hugging Face; OpenAI API; Google Collab; Librosa API)

github/ShadowKrius/Resonance

Awarded **Best Generative AI Hack – HackHer(413) 2025, sponsored by MLH**

- Created a pipeline that analyzes a song’s audio features (tempo, pitch, mood) and lyrics to generate a synchronized music video using AI.
- Combined audio analysis (Librosa) with prompt engineering to produce relevant visual scenes via OpenAI’s image generation, and compiled frames with OpenCV/FFmpeg; the project won “Best Generative AI Hack” for its creative approach.

SafeSteps (Flutter; Dart, Firebase, Google Maps API)

github/ShadowKrius/safesteps

Awarded **Best UMass Hack – HackUMass XII 2024**

- Developed a cross-platform safety app with real-time navigation to HELP phones and live location sharing for campus security.
- Engineered smart routing logic that prioritized safety by avoiding high-risk zones while minimizing travel delay, enhancing real-world applicability.
- Optimized app performance using GetX for state management and Firebase for seamless data sync.

TECHNICAL SKILLS

Languages: Python; Java; C; JavaScript; TypeScript; HTML/CSS; Dart;

Libraries: Pandas; TensorFlow; NumPy; Matplotlib; Scikit-learn; Qiskit;

Tools & Frameworks: FastAPI; Flask; Flutter; Firebase; Git; Docker; ReactJS; NextJS;

INTERESTS

Formula 1 racing, Japanese culture, Basketball, Reading Fiction & Non-fiction books, Cricket